CE 3372 Water Systems Design
Final Exam
Spring 2026

1. (10 points) What is a water use system? How are they usually sized?

2. (10 points) What is a water control system? How are they usually sized?

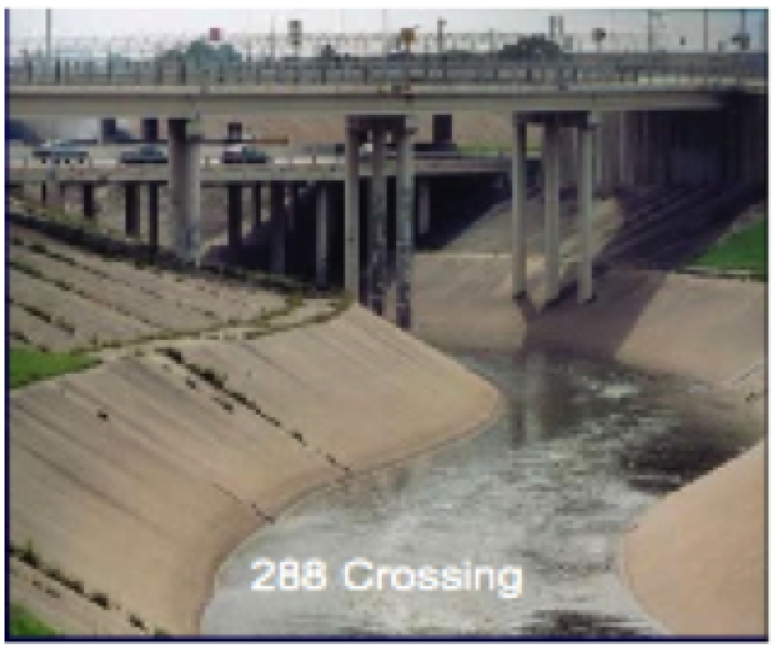


Figure 1: Braes Bayou at SH288

3. (10 points) Which kind of system is depicted in Figure 1? Explain your reasoning.

4. (10 points) Use Title 30, Part 1, Chapter 290, Subchapter D, Rule 290.44 of the Texas Administrative Code or TCEQ RG-195 to find the minimum pressure requirement for a public water system during normal (non-fire suppression) operation for parts of the system that deliver 1.5 gallons per minute or more.

5. (10 points) Use Title 30, Part 1, Chapter 290, Subchapter D, Rule 290.44 of the Texas Administrative Code or TCEQ RG-195 to find the minimum pressure requirement for a public water system during emergency (fire suppression) operation for parts of the system that deliver 1.5 gallons per minute or more.

6. (10 points) Use Title 30, Part 1, Chapter 290, Subchapter D, Rule 290.44 of the Texas Administrative Code or TCEQ RG-195 to find the minimum distance (spacing) for new potable water distribution lines in feet to wastewater collection facilities.

7. (10 points) Use Title 30, Part 1, Chapter 290, Subchapter D, Rule 290.44 of the Texas Administrative Code or TCEQ RG-195 to find the minimum free chlorine residual in mg/L required in Texas water distribution systems (if using free chlorine).

8. (30 points) Consider the two networks depicted in Figure 2. Both network designs serve the same junction nodes, have same nodal demands, and draw supply from the same reservoir.

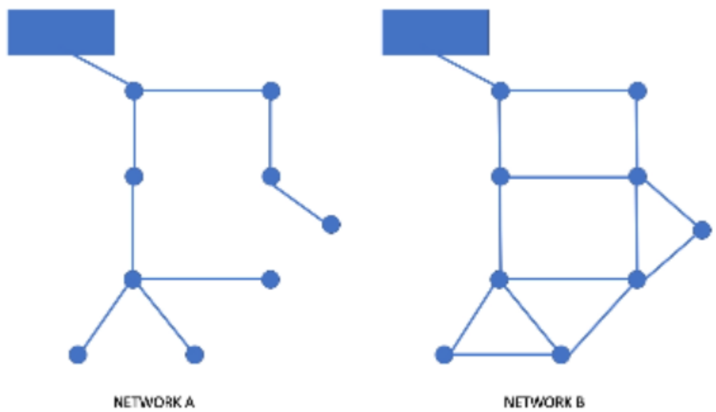


Figure 2: Two Distribution Systems

Determine:

a) Which network design would be superior in terms of reliability? Why?

b) Which network design would cost less to build? Why?

c) Which network would be superior in terms of disinfection residual? Why?

9. (30 points) Consider the pipe network portion shown in Figure 3. Node 6 has a total head of 290.5 meters. All the pipes are PVC (roughness height = 0.007 millimeters).

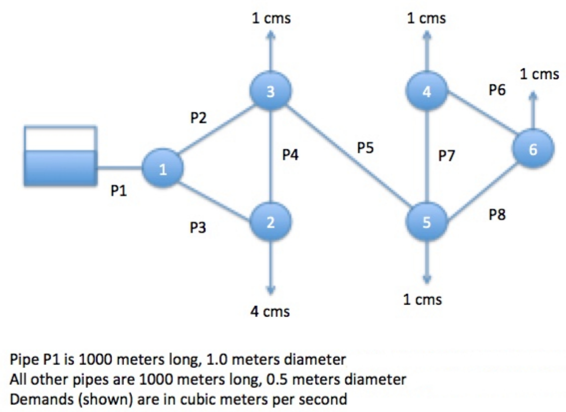


Figure 3: A Distribution System

Determine:

a) The discharge in cubic meters per second (cms) in pipe P1?

b) The discharge in cubic meters per second (cms) in pipe P5?

c) The discharge in cubic meters per second (cms) in pipe P6?

10. (10 points) Figure 4 is a perspective view of a large at-grade water storage reservoir. The initial depth in the 50-foot diameter tank is Z = 30 feet. What is the average outflow rate in cubic-feet-per-second, if the tank drains through the hole in the side in 24 hours?

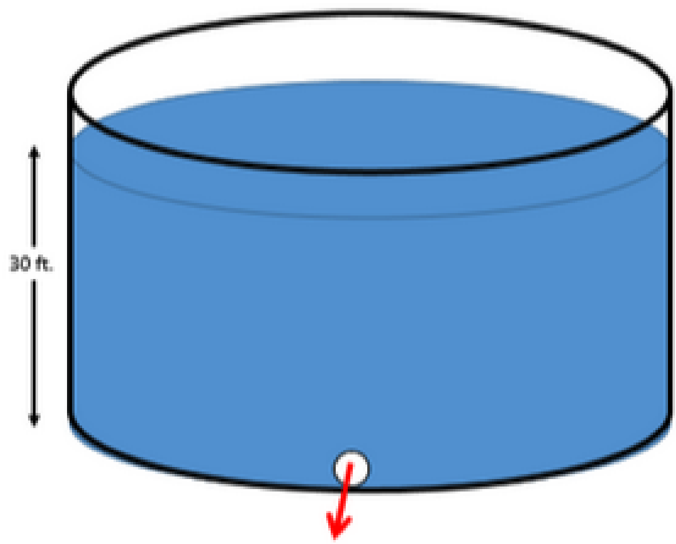


Figure 4: At-grade storage reservoir

11. (230 points) Figure 5 is a collection of annotated images of the Edmonston Pumping Plant and pipeline/tunnel system at the southern end of the California Aqueduct. The 10+ mile system lifts water from the Edmonston forebay at elevation 1239 feet to the Tehachapi afterbay at elevation 3131 feet for subsequent distribution to various parts of southern California.

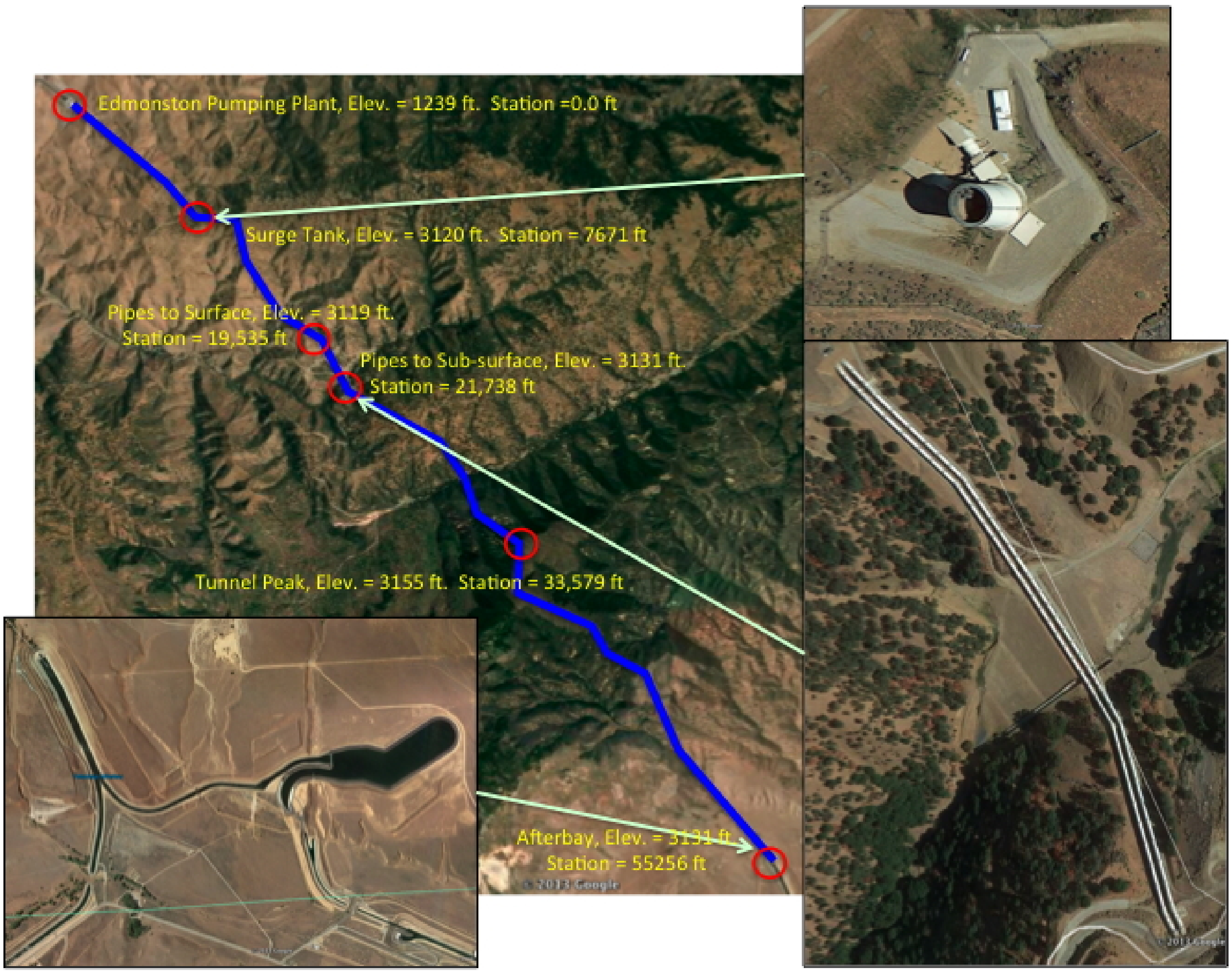


Figure 5: Aerial images of Tehachapi Mountains, California; showing the lift system for the California Aqueduct.

Figure 6 is a pump performance curve for any **one** of the 14 pumps arranged in two 7-pump bays, each supplying one of the parallel 16-foot diameter steel pressure pipes.

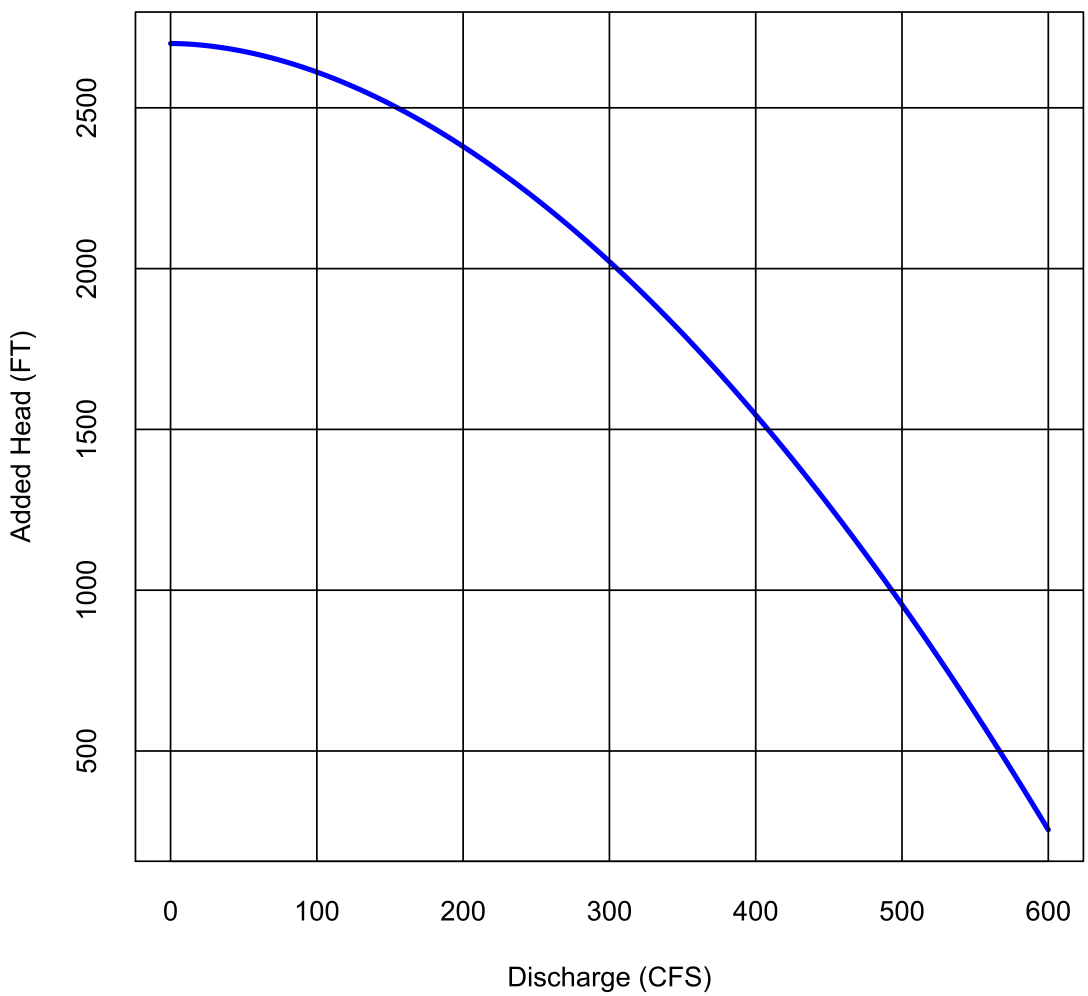


Figure 6: Lift pump performance curve for single pump. System has 14 identical pumps in two parallel galleries; each gallery has 7 pumps operating in parallel.

Determine:

a) A plan view layout of the system (a sketch) showing the various components, and station distances (from the pump station forebay). Label the sketch with node elevations, node names, pipe names, and pipe lengths.).

b) The pipe lengths between the indicated locations in Table 1 (i.e. complete the table)?

Table 1: Selected Pipeline Locations and Pressures

ID	Location	Distance from Pumps (feet)	Connection Sequence	Pipe Barrel Length
1	Edmonston Pump House	0	N/A	0
2	Surge Tank	7,671	1 to 2	
3	Pipes to Surface	19,535	2 to 3	
4	Pipes to Ground	21,738	3 to 4	
5	High Tunnel	33,579	4 to 5	
6	Tehachapi Afterbay	55,256	5 to 6	

c) The total length of pipe/tunnel required (in miles).

d) The total head at the Edmonston Pump Forebay (the pumphouse).

e) What hydraulic element to represent the forebay.

f) The total head at the Tehachapi Afterbay.

g) What hydraulic element to represent the afterbay.

h) The head-loss model is most appropriate for this analysis.

i) The total discharge in the system in cubic-feet-per-second when all 14 pumps are running.

j) The pressure at the locations in Table 2 (i.e. complete the table).

Table 2: Selected Pipeline Locations and Pressures

ID	Location	Distance from Pumps (feet)	Elevation (feet)	Pressure (psi)
0	Edmonston Pump House (suction)	0	1,239	
0	Edmonston Pump House (discharge)	0	1,239	
2	Surge Tank	7,671	3,121	
3	Pipes to Surface	19,535	3,119	
4	Pipes to Ground	21,738	3,131	
5	High Tunnel	33,579	3,566	
6	Tehachapi Afterbay	55,256	3,131	

k) The velocity of water the each of the pipes.

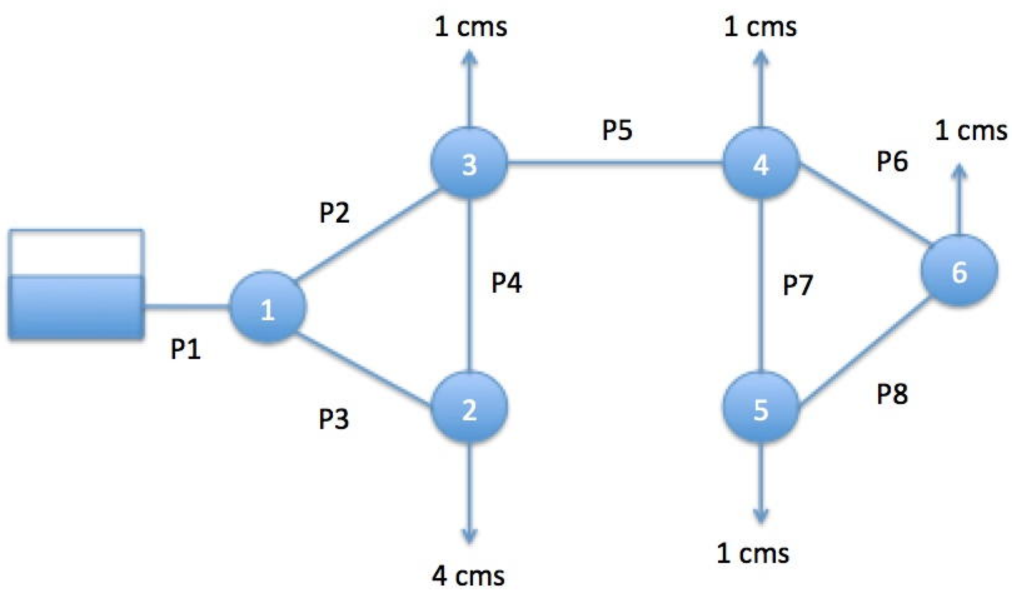
l) The mechanical energy required to move the water in kW-h (kilowatt-hours).

12. (46 Points) Consider the pipe network portion shown in Figure 7.

    The reservoir has a pool elevation of 1310 meters

    Node 1 elevation is 200.0 meters; Node 2 elevation is 150.0 meters
    Node 3 elevation is 150.0 meters; Node 4 elevation is 100.0 meters
    Node 5 elevation is 100.0 meters; Node 6 elevation is 50.0 meters

    P1 roughness is $e = 0.045$ mm; P2 roughness is $e = 0.045$ mm
    P3 roughness is $e = 0.045$ mm; P4 roughness is $e = 0.090$ mm
    P5 roughness is $e = 0.045$ mm; P6 roughness is $e = 0.045$ mm
    P7 roughness is $e = 0.045$ mm; P8 roughness is $e = 0.045$ mm



Pipe P1 is 1000 meters long, 1.0 meters diameter
All other pipes are 1000 meters long, 0.5 meters diameter
Demands (shown) are in cubic meters per second

Figure 7: Pipe Network

Using EPANET, determine:

a) The discharge, velocity, and head loss in each pipe and populate Table 9

Table 3: Estimated Q,V, and Δh in each pipe in Figure 7

Pipe ID	Discharge (m^3/s)	Velocity (m/s)	Head Loss (m)
1			
2			
3			
4			
5			
6			
7			
8			

b) The total head and pressure at each node and populate Table 4 below.

Table 4: TDH, z, and P at each node in Figure 7

Node ID	Total Head (m)	Elevation (m)	Pressure (kPa)
Reservoir	1310	N/A	0
1		200	
2		150	
3		150	
4		100	
5		100	
6		50	

c) Insert a **screen capture** of **your** EPANET simulation here.

d) Attach the EPANET **summary report** here.

13. (10 points) In a sewer system if design criteria require a minimum velocity of 2 ft/sec, and your design cannot produce that velocity at the design flow, what are some changes can you make in your design?

14. (10 points) Why is population forecasting important for water distribution and wastewater collection system designs?

15. (10 points) What is a lift station, and when are they necessary?

16. (10 points) Define the time of concentration for a small watershed.

17. (10 points) Define the inlet time for an urbanized drainage area.

18. (10 points) Define inflow and infiltraton in the context of a sanitary sewer system.

19. (10 points) A 24-inch diameter sewer pipe, with Manning's n of 0.015 is laid on slope $S_0 = 0.01$ as shown in Figure 8.

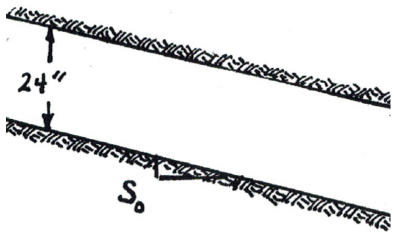


Figure 8: Sewer pipe sketch

Use Manning's equation and the depth-area, and the depth-perimeter equations to complete Table 5.

Table 5: Depth-Area, Depth-Perimeter, Depth-Hyd. Radius, and Discharge for Circular Sewer

y (ft)	A (ft^2)	P_w (ft)	R_h (ft)	Q (ft^3/sec)
1.00				
2.00				

20. (150 points)Consider the drainage area depicted in Figure 9. The 6.0 acre drainage area has a runoff coefficient of 0.65, and an inlet time (time of concentration to the inlet depicted in the figure) of 17 minutes. The pipes are concrete (Manning's n = 0.013). The initial invert elevations for the junction boxes (MH-1, MH-2, MH-3) are 323.2, 321.1, and 319.0 feet, respectively.

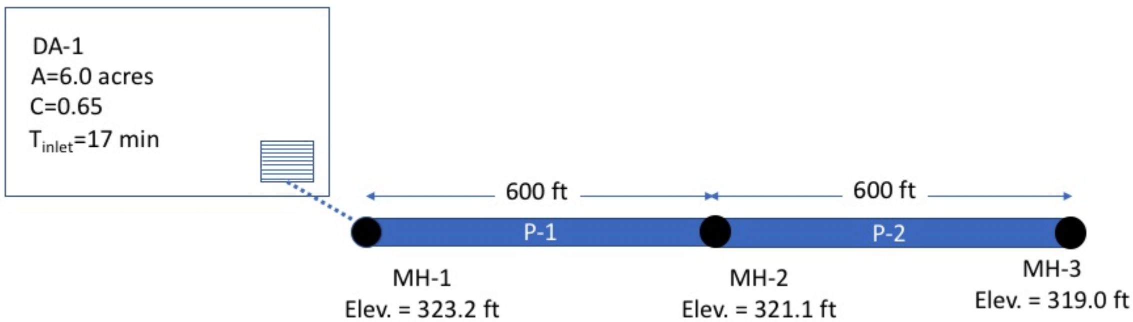


Figure 9: Drainage System Layout

Equation 1 is the 10-year ARI intensity equation for the area, where I is intensity in inches-per-hour, and T_c is the characteristic time, in minutes.

$$I = \frac{56.6}{(T_c + 8.6)^{0.823}} \tag{1}$$

Determine:

a) The dimensionless slope of the pipe run P1 (connecting MH-1 and MH-2)?

b) The dimensionless slope of the pipe run P2 (connecting MH-2 and MH-3)?

c) The CA value for drainage area DA-1?

d) The $\sum CA$ value (sum of the upstream products of runoff coefficients and contributing areas) at junction MH-1?

e) The $\sum CA$ value (sum of the upstream products of runoff coefficients and contributing areas) at junction MH-2?

f) The time of concentration to be used to compute the discharge leaving MH-1?

g) The rainfall intensity for this time of concentration?

h) The value of peak discharge leaving MH-1?

i) The pipe diameter for pipe P-1 required the carry the design flow at full pipe depth?

j) The flow velocity in pipe P-1?

k) The pipe travel time for pipe P-1?

l) The value of peak discharge leaving MH-2? (Explain your reasoning)

m) The pipe diameter for pipe P-2 required the carry the design flow at full pipe depth?

n) The flow velocity in pipe P-2?

o) The pipe travel time for pipe P-2?

21. (140 points) Consider the drainage area depicted in Figure 10.

Figure 10: Drainage System Layout

Table 6 lists the hydrologic features of the four drainage areas.

Table 6: Hydrologic data for each drainage area in Figure 10

Area ID	Area (*acres*)	Width (*feet*)	Avg. Slope (%)	NRCS CN
DA_1	25	1073	0.2	85
DA_2	25	1073	0.2	85
DA_3	17	719	0.8	85
DA_4	8	688	0.9	85

Table 7: Elevation data for each junction in Figure 10

Junction ID	Invert Elevation (*feet*)	Land Elevation (*feet*)
J-DA_1	65	70
J-DA_2	60	66
J-DA_3	56	66
J-DA_4	48	62
OUTFALL	35	N/A

The red dots on Figure 10 are inlets to the storm sewer system (indicated by the blue pipes). The red rectangle (on the East side of the figure) is a stormwater outfall. Table 7 lists the invert elevations, and land surface elevations of these points.

The system is to be sized using the design storm shown in Table 8[1]

Table 8: Design Storm for Drainage System in Figure 10

Elapsed Time	Rainfall Intensity ($\frac{in}{hr}$)
00:00	0.11
01:00	0.11
02:00	0.13
03:00	0.13
04:00	0.15
05:00	0.17
06:00	0.19
07:00	0.21
08:00	0.27
09:00	0.34
10:00	0.54
11:00	4.28
12:00	1.09
13:00	0.48
14:00	0.34
15:00	0.26
16:00	0.22
17:00	0.19
18:00	0.17
19:00	0.14
20:00	0.13
21:00	0.12
22:00	0.12
23:00	0.11

The blue line segments on Figure 10 are pipes that comprise the storm sewer system. Table 9 lists the proposed design (i.e. pipe lengths, diameters, connection(s), and offsets) for the storm sewer.

[1] A 10-inch, 24 hour, SCS Type-2 - with the first and last hour trimmed

Table 9: Hydraulic data for each pipe in Figure 10

Pipe ID	Connection (Start:End)	Diameter $(feet)$	Length $(feet)$	Offset (inlet:outlet)
P_1	DA_1 : DA_2	3.0	1000	0:0
P_2	DA_2 : DA_4	3.0	1000	0:2
P_3	DA_3 : DA_4	3.0	1000	0:2
P_4	DA_4 : OUTFALL	5.0	1000	0:0

Using these data build a SWMM model of the system. Assume the outfall is a fixed pool with depth 5 feet (water elevation 40 feet). Use DYNWAVE routing, and the SCS CN infiltration method in SWMM.

From **your** SWMM model, determine:

a) Insert a screen capture of your working model here.

b) How many subcatchments are in your model?

c) If all the pipes are below grade?

d) If the system surcharges during the design storm?

e) If yes, when does the surcharge occur (what time)?

f) The maximum flow velocity in the sewer system?

g) Where and when this maximum occurs (which pipe, what time)?

h) Insert a profile of the sewer system and the water surface elevation at hour 12:00 of the simulation, starting at J-DA_1 and ending at the OUTFALL

i) Insert a time series (a plot) of the precipitation input

j) Insert a time series (a plot) of runoff in subcatchment DA_2

Modify your model to make pipes P_1 and P_2, into 5.0 foot diameter pipes, keeping them at least 2 feet below grade.

k) List the required changes (there are several, invert elevations must be changed, and offsets need changing)

l) Does the system surcharge during the design storm?

m) Insert a profile of the **modified** sewer system and the water surface elevation at hour 12:00 of the simulation, starting at J-DA_1 and ending at the OUTFALL

n) Insert a time series (a plot) of runoff in subcatchment DA_2